

Kushagra

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Education

Vellore Institute of Technology

B.Tech in Computer Science Engineering (Cybersecurity & Digital Forensics)

2022 – 2026

CGPA: 7.55

Technical Skills

Languages: Java (Primary), Python, JavaScript

Backend: Spring Boot, REST API Design, MVC Architecture

Database: MySQL (Query Optimization, CRUD Operations)

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, CI/CD

Testing: JUnit

Tools: Git, GitHub

Projects

AgroVita – ML-Integrated Backend Deployment Platform

Java, Spring Boot, Python (ML), MySQL, Docker, AWS

- Developed scalable REST APIs using Spring Boot to integrate ML-based crop prediction for agricultural automation
- Designed layered architecture (Controller–Service–Repository) improving maintainability and reducing code coupling
- Containerized application using Docker ensuring consistent deployment across environments
- Deployed backend on AWS EC2 and implemented IAM-based access control for secure system operations
- Integrated AWS S3 for efficient storage of datasets and deployment artifacts
- Built CI/CD pipeline reducing manual deployment effort and improving delivery efficiency

Weather Forecast Backend System

Spring Boot, MySQL, JUnit

- Built RESTful backend services integrating third-party weather APIs with optimized response handling
- Implemented global exception handling improving API reliability and error transparency
- Designed clean MVC architecture ensuring modular and scalable backend structure
- Developed unit tests using JUnit improving code reliability and reducing bugs
- Managed MySQL database for persistent logging and data tracking

Secure VPN Configuration & Network Analysis

Linux, VPN Protocols

- Configured secure VPN communication using tunneling protocols ensuring encrypted data transmission
- Analyzed IPsec and OpenVPN protocols to evaluate performance and security trade-offs
- Applied cybersecurity principles to identify potential vulnerabilities in secure communication systems

Smart Contract Development (Ongoing)

Solidity, Ethereum

- Developing Ethereum smart contracts focusing on gas optimization and secure state transitions
- Studying EVM execution model and analyzing vulnerabilities such as reentrancy attacks

Certifications

Oracle Java Foundation Associate (2025)

IBM Cybersecurity Analyst (2025)

JPMorgan Chase Software Engineering Job Simulation (2026)

Achievements

- Solved 1400+ Data Structures & Algorithms problems on LeetCode
- Achieved Global Rank 7,000 on LeetCode

Key Strengths

- Strong problem-solving skills with extensive practice in Data Structures and Algorithms
- Solid understanding of backend system design, REST APIs, and scalable architecture
- Working knowledge of cybersecurity principles including secure communication and vulnerability awareness
- Experience deploying and managing applications using AWS and Docker